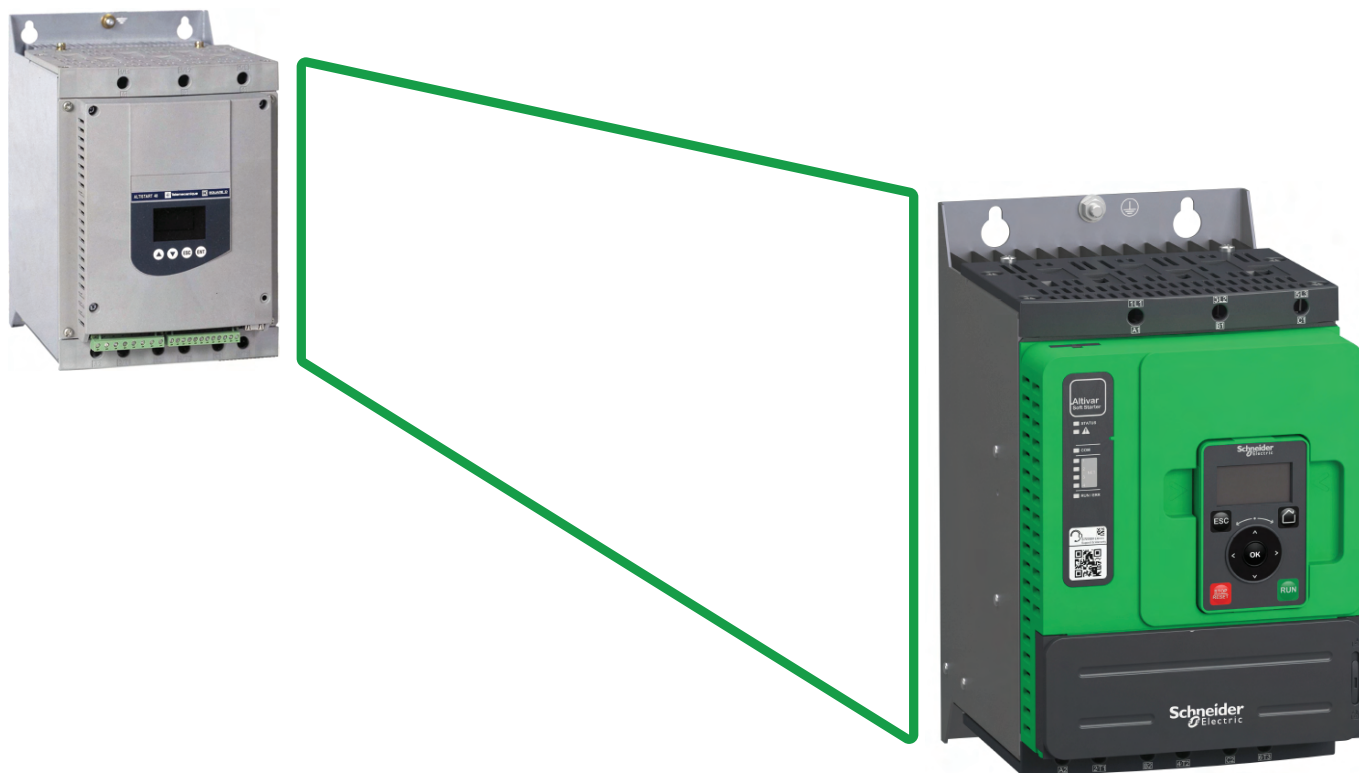


Altivar Soft Starter ATS480

ATS48 to ATS480 Substitution Manual

NNZ85529.02
05/2025



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The information provided in this document contains general descriptions, technical characteristics and/or recommendations related to products/solutions.

This document is not intended as a substitute for a detailed study or operational and site-specific development or schematic plan. It is not to be used for determining suitability or reliability of the products/solutions for specific user applications. It is the duty of any such user to perform or have any professional expert of its choice (integrator, specifier or the like) perform the appropriate and comprehensive risk analysis, evaluation and testing of the products/solutions with respect to the relevant specific application or use thereof.

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Safety Information

Important Information

Read these instructions carefully, and look at the equipment to become familiar with the device before trying to install, operate, service, or maintain it. The following special messages may appear throughout this documentation or on the equipment to warn of potential hazards or to call attention to information that clarifies or simplifies a procedure.



The addition of this symbol to a “Danger” or “Warning” safety label indicates that an electrical hazard exists which will result in personal injury if the instructions are not followed.



This is the safety alert symbol. It is used to alert you to potential personal injury hazards. Obey all safety messages that follow this symbol to avoid possible injury or death.

DANGER

DANGER indicates a hazardous situation which, if not avoided, **will result in** death or serious injury.

WARNING

WARNING indicates a hazardous situation which, if not avoided, **could result in** death or serious injury.

CAUTION

CAUTION indicates a hazardous situation which, if not avoided, **could result in** minor or moderate injury.

NOTICE

NOTICE is used to address practices not related to physical injury.

Please Note

Electrical equipment should be installed, operated, serviced, and maintained only by qualified personnel. No responsibility is assumed by Schneider Electric for any consequences arising out of the use of this material.

A qualified person is one who has skills and knowledge related to the construction and operation of electrical equipment and its installation, and has received safety training to recognize and avoid the hazards involved.

Qualification of Personnel

Only appropriately trained persons who are familiar with and understand the contents of this manual and all other pertinent product documentation are authorized to work on and with this product. In addition, these persons must have received safety training to recognize and avoid hazards involved. These persons must have sufficient technical training, knowledge and experience and be able to foresee and detect potential hazards that may be caused by using the product, by changing the settings and by the mechanical, electrical and electronic equipment of the entire system in which the product is used. All persons working on and with the product must be fully familiar with all applicable standards, directives, and accident prevention regulations when performing such work.

Intended Use

This product is intended for industrial use according to this manual.

The product may only be used in compliance with all applicable safety standard and local regulations and directives, the specified requirements and the technical data. The product must be installed outside the hazardous ATEX zone. Prior to using the product, you must perform a risk assessment in view of the planned application. Based on the results, the appropriate safety measures must be implemented. Since the product is used as a component in an entire system, you must ensure the safety of persons by means of the design of this entire system (for example, machine design). Any use other than the use explicitly permitted is prohibited and can result in hazards.

About the document

Document Scope

The purpose of this document is to:

- Show you how to the Modbus RTU fieldbus on your soft starter.
- Show you how to configure the soft starter to use the Modbus RTU for monitoring and control.
- Provide examples of setup using Modbus RTU communication.

NOTE: Read and understand this document and all related documents (see below) before installing, operating, or maintaining your soft starter.

Validity Note

Original instructions and information given in the present document have been written in English (before optional translation).

NOTE: The products listed in the document are not all available at the time of publication of this document online. The data, illustrations and product specifications listed in the guide will be completed and updated as the product availabilities evolve. Updates to the guide will be available for download once products are released onto the market.

This documentation is valid only for ATS480.

The characteristics of the products described in this document are intended to match the characteristics that are available on www.se.com. As part of our corporate strategy for constant improvement, we may revise the content over time to enhance clarity and accuracy. If you see a difference between the characteristics in this document and the characteristics on www.se.com, consider www.se.com to contain the latest information.

Product Related Information

Read and understand these instructions before performing any procedure with this soft starter.

DANGER

HAZARD OF ELECTRIC SHOCK, EXPLOSION, OR ARC FLASH

- Only appropriately trained persons who are familiar with and fully understand the contents of the present manual and all other pertinent product documentation and who have received all necessary training to recognize and avoid hazards involved are authorized to work on and with this equipment.
- Installation, adjustment, repair and maintenance must be performed by qualified personnel.
- Verify compliance with all local and national electrical code requirements as well as all other applicable regulations with respect to grounding of all equipment.
- Only use properly rated, electrically insulated tools and measuring equipment.
- Do not touch unshielded components or terminals with voltage present.
- Prior to performing any type of work on the equipment, block the motor shaft to prevent rotation.
- Insulate both ends of unused conductors of the motor cable.

Failure to follow these instructions will result in death or serious injury.

DANGER

HAZARD OF ELECTRIC SHOCK, EXPLOSION, OR ARC FLASH

Before performing work on the equipment:

- Use all required personal protective equipment (PPE).
- Disconnect all power, including external control power that may be present. Take into account that the circuit breaker or main switch does not de-energize all circuits.
- Place a "Do Not Turn On" label on all power switches related to the equipment.
- Lock all power switches in the open position.
- Verify the absence of voltage using a properly rated voltage sensing device.

Before applying voltage to the equipment:

- Verify that the work has been completed and that the entire installation cannot cause hazards.
- If the mains input terminals and the motor output terminals have been grounded and short-circuited, remove the ground and the short circuits on the mains input terminals and the motor output terminals.
- Verify proper grounding of all equipment.
- Verify that all protective equipment such as covers, doors, grids is installed and/or closed.

Failure to follow these instructions will result in death or serious injury.

DANGER

HAZARD OF ELECTRIC SHOCK, EXPLOSION, OR ARC FLASH

- Never operate energized switch with door open.
- Turn off switch before removing or installing fuses or making load side connections.
- Do not use renewable link fuses in fused switches.

Failure to follow these instructions will result in death or serious injury.

Damaged products or accessories may cause electric shock or unanticipated equipment operation.

DANGER

ELECTRIC SHOCK OR UNANTICIPATED EQUIPMENT OPERATION

Do not use damaged products or accessories.

Failure to follow these instructions will result in death or serious injury.

Contact your local Schneider Electric sales office if you detect any damage whatsoever.

This equipment has been designed to operate outside of any hazardous location. Only install this equipment in zones known to be free of a hazardous atmosphere.

DANGER

POTENTIAL FOR EXPLOSION

Install and use this equipment in non-hazardous locations only.

Failure to follow these instructions will result in death or serious injury.

Your application consists of a whole range of different interrelated mechanical, electrical, and electronic components, the device being just one part of the application. The device by itself is neither intended to nor capable of providing the entire functionality to meet all safety-related requirements that apply to your application. Depending on the application and the corresponding risk assessment to be conducted by you, a whole variety of additional equipment is required such as, but not limited to, external monitoring devices, guards, etc.

As a designer/manufacturer of machines, you must be familiar with and observe all standards that apply to your machine. You must conduct a risk assessment and determine the appropriate Performance Level (PL) and/or Safety Integrity Level (SIL) and design and build your machine in compliance with all applicable standards. In doing so, you must consider the interrelation of all components of the machine. In addition, you must provide instructions for use that enable the user of your machine to perform any type of work on and with the machine such as operation and maintenance in a safe manner.

The present document assumes that you are fully aware of all normative standards and requirements that apply to your application. Since the device cannot provide all safety-related functionality for your entire application, you must ensure that the required Performance Level and/or Safety Integrity Level is reached by installing all necessary additional equipment.

⚠ WARNING

INSUFFICIENT PERFORMANCE LEVEL/SAFETY INTEGRITY LEVEL AND/OR UNINTENDED EQUIPMENT OPERATION

- Conduct a risk assessment according to EN ISO 12100 and all other standards that apply to your application.
- Use redundant components and/or control paths for all critical control functions identified in your risk assessment.
- Verify that the service life of all individual components used in your application is sufficient for the intended service life of your overall application.
- Perform extensive commissioning tests for all potential error situations to verify the effectiveness of the safety-related functions and monitoring functions implemented, for example, but not limited to, speed monitoring by means of encoders, short circuit monitoring for all connected equipment, correct operation of brakes and guards.
- Perform extensive commissioning tests for all potential error situations to verify that the load can be brought to a safe stop under all conditions.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

Product may perform unexpected movements because of incorrect wiring, incorrect settings, incorrect data or other errors.

⚠ WARNING

UNANTICIPATED EQUIPMENT OPERATION

- Carefully install the wiring in accordance with the EMC requirements.
- Do not operate the product with unknown or unsuitable settings or data.
- Perform a comprehensive commissioning test.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

⚠ WARNING

LOSS OF CONTROL

- The designer of any control scheme must consider the potential failure modes of control paths and, for critical control functions, provide a means to achieve a safe state during and after a path failure. Examples of critical control functions are emergency stop, overtravel stop, power outage and restart.
- Separate or redundant control paths must be provided for critical control functions.
- System control paths may include communication links. Consideration must be given to the implications of unanticipated transmission delays or failures of the link.
- Observe all accident prevention regulations and local safety guidelines (1).
- Each implementation of the product must be individually and thoroughly tested for proper operation before being placed into service.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

(1) For USA: Additional information, refer to NEMA ICS 1.1 (latest edition), Safety Guidelines for the Application, Installation and Maintenance of Solid State Control.

Machines, controllers, and related equipment are usually integrated into networks. Unauthorized persons and malware may gain access to the machine as well as to other devices on the network/fieldbus of the machine and connected networks via insufficiently secure access to software and networks.

⚠ WARNING

UNAUTHORIZED ACCESS TO THE MACHINE VIA SOFTWARE AND NETWORKS

- In your hazard and risk analysis, consider all hazards that result from access to and operation on the network/fieldbus and develop an appropriate cyber security concept.
- Verify that the hardware infrastructure and the software infrastructure into which the machine is integrated as well as all organizational measures and rules covering access to this infrastructure consider the results of the hazard and risk analysis and are implemented according to best practices and standards covering IT security and cyber security (such as: ISO/IEC 27000 series, Common Criteria for Information Technology Security Evaluation, ISO/IEC 15408, IEC 62351, ISA/IEC 62443, NIST Cybersecurity Framework, Information Security Forum - Standard of Good Practice for Information Security, SE recommended Cybersecurity Best Practices*).
- Verify the effectiveness of your IT security and cyber security systems using appropriate, proven methods.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

(*) : SE Recommended Cybersecurity Best Practices can be downloaded on [SE.com](https://se.com).

⚠ WARNING

LOSS OF CONTROL

Perform a comprehensive commissioning test to verify that communication monitoring properly detects communication interruptions.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

This product meets the EMC requirements according to the standard IEC 60947-4-2. This device has been designed for environment A. Use of this product in a domestic environment (B environment) may cause unwanted radio interference.

⚠ WARNING

RADIO INTERFERENCE

- In a domestic environment (B environment), this product may cause radio interference in which case supplementary mitigation measures may be required.
- The references from ATS480D17Y to ATS480C11Y can be adapted to a domestic environment (B environment) by adding an external bypass contactor. For other ATS480 references, you must consider other mitigation measures.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

NOTICE

DESTRUCTION DUE TO INCORRECT MAINS VOLTAGE

Before switching on and configuring the product, verify that it is approved for the mains voltage.

Failure to follow these instructions can result in equipment damage.

General Cybersecurity Information

In recent years, the growing number of networked machines and production plants has seen a corresponding increase in the potential for cyber threats, such as unauthorized access, data breaches, and operational disruptions. You must, therefore, consider all possible cybersecurity measures to help protect assets and systems against such threats.

To help keep your Schneider Electric products secure and protected, it is in your best interest to implement the cybersecurity best practices as described in the [Cybersecurity Best Practices](#) document.

Schneider Electric provides additional information and assistance:

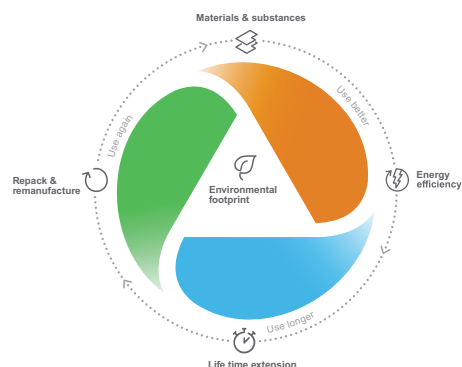
- Subscribe to the Schneider Electric [security newsletter](#).
- Visit the [Cybersecurity Support Portal](#) web page to:
 - Find Security Notifications.
 - Report vulnerabilities and incidents.
- Visit the [Schneider Electric Cybersecurity and Data Protection Posture](#) web page to:
 - Access the cybersecurity posture.
 - Learn more about cybersecurity in the cybersecurity academy.
 - Explore the cybersecurity services from Schneider Electric.

Environmental Data

The Environmental Data Program is a framework for how we measure, categorize, and compare the environmental attributes and footprint of our products.

Using a rigorous, fact-based methodology, the program provides environmental data from across the product lifecycle.

Five data categories across the product lifecycle



Use Better: How sustainable a product is, including environmental footprint, materials and substances, packaging, and energy efficiency.

Use Longer: How a product's life time can be effectively extended in terms of reparability and updatability.

Use Again: How a product can be reused, from dismantling and remanufacturing to recyclability and manufacturer take back.

With this transparent, verified data, customers and partners are empowered to make conscious environmental choices and accurately evaluate and report on sustainability performance.

All our hardware offers have an associated environmental data available on se.com product pages.

Refer to Environmental Data Program for more information.

Related Documents

Use your tablet or your PC to quickly access detailed and comprehensive information on all our products on www.se.com The Internet site provides the information you need for products and solutions:

- The whole catalog for detailed characteristics and selection guides
- The CAD files to help design your installation, available in over 20 different file formats
- All software and firmware to maintain your installation up to date
- A large quantity of White Papers, Environment documents, Application solutions, Specifications... to gain a better understanding of our electrical systems and equipment or automation
- And finally all the User Guides related to your soft starter, listed below:

You can download there technical publications and other technical information from our website at www.se.com/en/download.

Catalog

Title of documentation	Reference number
Catalog: Altivar Soft Starter ATS480	DIA2ED2210602EN (English) DIA2ED2210602FR (French) ECATA1172 (Chinese) DIA2ED2210602DE (German)

Documentations

Title of documentation	Reference number
ATS480 Getting Started Manual	NNZ85504 (English), NNZ85505 (French) NNZ85506 (Spanish), NNZ85507 (Italian) NNZ85508 (German), NNZ85509 (Chinese) NNZ85510 (Portuguese), NNZ85511 (Turkish)
ATS480 Getting Started Manual Annex for UL	NNZ86539 (English)
ATS480 User Manual	NNZ85515 (English), NNZ85516 (French) NNZ85517 (Spanish), NNZ85518 (Italian) NNZ85519 (German), NNZ85520 (Chinese) NNZ85521 (Portuguese), NNZ85522 (Turkish)
ATS480 Embedded Modbus RTU Manual	NNZ85539 (English)
ATS480 EtherNet/IP – Modbus TCP Manual	NNZ85540 (English)
ATS480 PROFIBUS DP Manual	NNZ85542 (English)
ATS480 Profinet Manual	NNZ85541 (English)
ATS480 CANopen Manual	NNZ85543 (English)
ATS480 Communication Parameter Addresses	NNZ85544 (English)
ATS480 Cascade Function Application Note	NNZ85564 (English)
Recommended Cybersecurity Best Practices	CS-Best-Practices-2019–340 (English)

ATS48 to ATS480 substitution

Title of documentation	Reference number
ATS48 to ATS480 Substitution Manual	NNZ85529 (English), NNZ85530 (French) NNZ85531 (Spanish), NNZ85532 (Italian) NNZ85533 (German), NNZ85534 (Chinese) NNZ85535 (Portuguese), NNZ85536 (Turkish)
Video: How to substitute an ATS48 for an ATS480?	FAQ000210049 (English)

Videos

Title of documentation	Reference number
Getting Started with ATS480	FAQ000233342 (English)
How to update the firmware on ATS480 with EcoStruxure Automation Device Maintenance?	FAQ000233943 (English)
How to configure the cybersecurity applied to ATS480?	FAQ000236206 (English)
How to Integrate ATS480 DFB with EcoStruxure Control Expert	FAQ000244312 (English)

Software

Title of documentation	Reference number
SoMove: FDT	SoMove FDT (English, French, German, Spanish, Italian, Chinese)
ATS480: DTM	ATS480 DTM Library EN (English – to be installed first) ATS480 DTM Lang FR (French) ATS480 DTM Lang SP (Spanish) ATS480 DTM Lang IT (Italian) ATS480 DTM Lang DE (German) ATS480 DTM Lang CN (Chinese)
EcoStruxure Automation Device Maintenance	EADM (English)

Information on Non-Inclusive or Insensitive Terminology

As a responsible, inclusive company, Schneider Electric is constantly updating its communications and products that contain non-inclusive or insensitive terminology. However, despite these efforts, our content may still contain terms that are deemed inappropriate by some customers.

Terminology used in this document

The technical terms, terminology, and the corresponding descriptions in this manual normally use the terms or definitions in the relevant standards.

Among others, these standards include:

- ISO 13849: The Foundation of Functional Safety in the Machinery
- IEC 60204-1: Safety of machinery - Electrical equipment of machines – Part 1: General requirements.
- IEC 61158 series: Industrial communication networks - Fieldbus specifications
- IEC 61508 Ed.2 series: Functional safety of electrical/electronic/programmable electronic safety-related.
- IEC 61784 series: Industrial communication networks - Profiles.
- IEC 61800 series: Adjustable speed electrical power drive systems.
- IEC 62443: Security for industrial automation and control systems.

In the area of drive systems this includes, but is not limited to, terms such as **error**, **error message**, **failure**, **fault**, **fault reset**, **protection**, **safe state**, **safety function**, **warning**, **warning message**, and so on.

In addition, the term **zone of operation** is used in conjunction with the description of specific hazards, and is defined as it is for a **hazard zone** or **danger zone** in the EC Machinery Directive (2006/42/EC) and in ISO 12100-1.

Structure of the Parameter Table

Pictogram	Description
	This parameter can be set during operation or when stopped. NOTE: It is advisable to stop the motor before modifying any of the settings
	The motor must be stopped to set this parameter.
	Power cycle must be performed after setting this parameter.
	Read only parameter, mainly used for monitoring.
	Expert mode required to access this parameter.

Contact us

Select your country on www.se.com/contact.

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How to substitute an ATS48 with an ATS480?

In complement of this manual a video presenting the substitution procedure is available on the Schneider Electric FAQ (FAQ000210049).



Substitution Procedure Overview

Step	Action
1	Make the inventory of the existing ATS48 installation: - Note the ATS48 commercial reference to replace by the ATS480 - Note the installed ATS48 options and accessories: - Remote display terminal VW3G48101 - Protective covers for power terminals - Line choke - DNV kits For more information refer to Inventory of the Existing ATS48 Installation, page 19.
2	Select the ATS480 commercial reference to replace the ATS48: - Select the soft starter reference with the ATS480 catalog or selector on se.com - Select the display terminal and remote mounting kit to match the required IP protection degree - Keep the existing protective covers for power terminals, line chokes and DNV kits For more information refer to ATS480 Soft Starter Selection and Accessories, page 22.
3	Uninstall the ATS48 and install the ATS480 in the same environment: - Same enclosure - Same clearances - Consider the depth difference between ATS48 and ATS480 - Install the new display terminal and the remote mounting kit if used For more information refer to Installation, page 28.
4	Wire the ground, supply mains and motor supply of the newly installed ATS480: - Reuse the cables from the ATS48 - Same cables section, length and position For more information refer to Power and Ground Wiring, page 37.
5	Wire the control terminals of the newly installed ATS480: - Reuse the cables from the ATS48 - Refer to the correspondence table between ATS48 and ATS480 control terminals NOTE: The ATS480 must be supplied with 110...230 Vac only For more information refer to Layout And Characteristics Of Control Terminals, page 38.
6	Before switching On refer to Checking Installation, page 42.
7	Initialize the ATS480: - Set the language and date - Set the cybersecurity policy of the ATS480 This menu is displayed when the ATS480 is supplied for the first time. For more information refer to Layout And Characteristics Of Control Terminals, page 38.
8	Migrate the ATS48 configuration to the newly installed ATS480: - The ATS480 must be supplied - Use the migration tool from SoMove For more information refer to Configuration Migration, page 45. This concludes the substitution from the ATS48 to the ATS480.

Inventory of the Existing ATS48 Installation

What’s in This Part

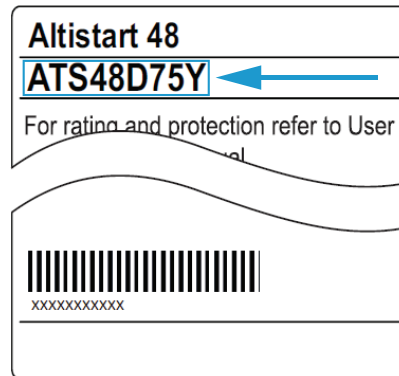
Identification of the ATS48 Commercial Reference.....20

Identification of the Accessories and Options Commercial References21

Identification of the ATS48 Commercial Reference

Identify the commercial reference of the ATS48 to substitute with the ATS480.

The ATS48 reference can be found on its nameplate label localized on the right side or on the front of the product:



Identification of the Accessories and Options Commercial References

Use this table to write down each ATS48 and related options to substitute in the existing installation.

	Quantities	ATS reference	Remote terminal VW3G48101 (Yes/No)	Protective covers power terminals (Yes/No)	Line chokes (Yes/No)	DNV kits (Yes/No)	Comments
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
11							
12							
13							
14							
15							
16							
17							

ATS480 Soft Starter Selection and Accessories

What's in This Part

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Line Chokes.....	26
DNV Kits.....	27

Soft Starter Selection

Refer to the following table to select the ATS480 compatible with the ATS48 to substitute.

Legacy offers			New offers
ATS48●●●Q 230...415 Vac	ATS48●●●Y 208...690 Vac		ATS480●●●Y 208...690 Vac
ATS48D17Q	ATS48D17Y		ATS480D17Y
ATS48D22Q	ATS48D22Y		ATS480D22Y
ATS48D32Q	ATS48D32Y		ATS480D32Y
ATS48D38Q	ATS48D38Y		ATS480D38Y
ATS48D47Q	ATS48D47Y		ATS480D47Y
ATS48D62Q	ATS48D62Y		ATS480D62Y
ATS48D75Q	ATS48D75Y		ATS480D75Y
ATS48D88Q	ATS48D88Y		ATS480D88Y
ATS48C11Q	ATS48C11Y		ATS480C11Y
ATS48C14Q	ATS48C14Y		ATS480C14Y
ATS48C17Q	ATS48C17Y		ATS480C17Y
ATS48C21Q	ATS48C21Y		ATS480C21Y
ATS48C25Q	ATS48C25Y		ATS480C25Y
ATS48C32Q	ATS48C32Y		ATS480C32Y
ATS48C41Q	ATS48C41Y		ATS480C41Y
ATS48C48Q	ATS48C48Y		ATS480C48Y
ATS48C59Q	ATS48C59Y		ATS480C59Y
ATS48C66Q	ATS48C66Y		ATS480C66Y
ATS48C79Q	ATS48C79Y		ATS480C79Y
ATS48M10Q	ATS48M10Y		ATS480M10Y
ATS48M12Q	ATS48M12Y		ATS480M12Y

The specific legacy references and their functionality are included in the substitution:

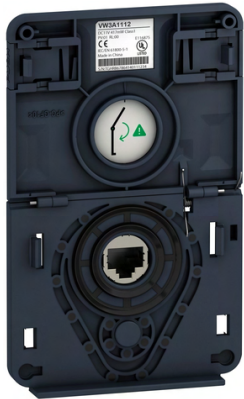
Specific legacy offers		New offers
ATS48●●●YS316, ability to set in Delta up to 500 V		ATS480●●●Y
ATS48●●●YS338, tropicalization coating		
ATS48●●●QS338, tropicalization coating		

Display Terminal and Remote Mounting Kit Selection



- The ATS48 remote terminal is not compatible with the ATS480. You cannot reuse it.
- To reach IP65 or higher protection degree, use the Graphic Display Terminal and its door mounting kit.
- For IP43, use the Plain Text Display Terminal and its door mounting kit.

Refer to the following table to choose a display terminal and its door mounting kit.

Door Mounting Kit protection degree	Display terminal	Door mounting kit
IP65	VW3A1111 graphic display terminal Available as option 	VW3A1112 remote mounting kit Available as option  Refer to the instruction sheet EAV7640603.
IP43	VW3A1113 plain text display terminal Included with the product 	VW3A1114 remote mounting kit Available as option  Refer to the instruction sheet EAV91355.
Use the cable VW3A1104R30 to replace the ATS48 3 meters cable for remote mounting kit. Not included with the remote kit		

Protective Covers for Power Terminals



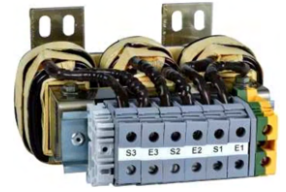
The ATS48 protective covers for power terminals are fully compatible with the ATS480 and can be reused.



Line Chokes



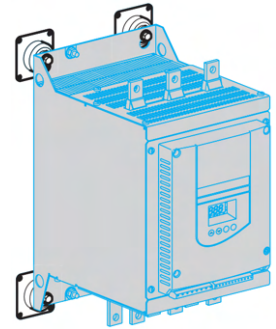
The ATS48 line chokes are fully compatible with the ATS480 and can be reused.



DNV Kits



- The ATS48 DNV kits are fully compatible with the ATS480 for the references from ATS480D17Y to ATS480C66Y
- For the references from ATS480C79Y to ATS480M12Y please refer to the ATS480 catalogue on SE.com to order the new kit



Installation

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Product related information

Conductive foreign objects may cause parasitic voltage.

 **DANGER**

ELECTRIC SHOCK AND/OR UNANTICIPATED EQUIPMENT OPERATION

- Keep foreign objects such as chips, screws or wire clippings from getting into the product.
- Verify correct seat of seals and cable entries in order to avoid deposits and humidity.

Failure to follow these instructions will result in death or serious injury.

The temperature of the products described in this manual may exceed 80 °C (176 °F) during operation.

 **WARNING**

HOT SURFACES

- Ensure that any contact with hot surfaces is avoided.
- Do not allow flammable or heat-sensitive parts in the immediate vicinity of hot surfaces.
- Verify that the product has sufficiently cooled down before handling it.
- Verify that the heat dissipation is sufficient by performing a test run under maximum load conditions.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

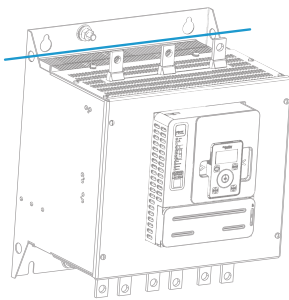
Handling



- The ATS480 and ATS48 must be handled the same way according to the following instructions.
- The ATS480 and ATS48 weights are identical.

Weight And Lifting Lugs Availability

See the weights, lifting lugs availability and packaging types in the following table before installing the soft starter.



References	Weight kg (lbs)	Lifting lugs	Packaging
ATS480D17Y...D47Y	4.9 (10.8)	No	Cardboard box
ATS480D62Y...C11Y	8.3 (18.2)	No	Cardboard box
ATS480C14Y...C17Y	12.4 (27.3)	Yes	Cardboard box
ATS480C21Y...C32Y	18.2 (40.1)	Yes	Pallet
ATS480C41Y...C66Y	51.4 (113.3)	Yes	Pallet
ATS480C79Y...M12Y	115 (253.5)	Yes	Pallet

Unpacking and Hoisting the References on Pallet

The references from ATS480C21Y to ATS480M12Y are mounted on pallet.

⚠ CAUTION

SHARP EDGES

Use all necessary personal protective equipment (PPE) such as gloves when removing the components from the pallet.

Failure to follow these instructions can result in injury or equipment damage.

⚠ WARNING

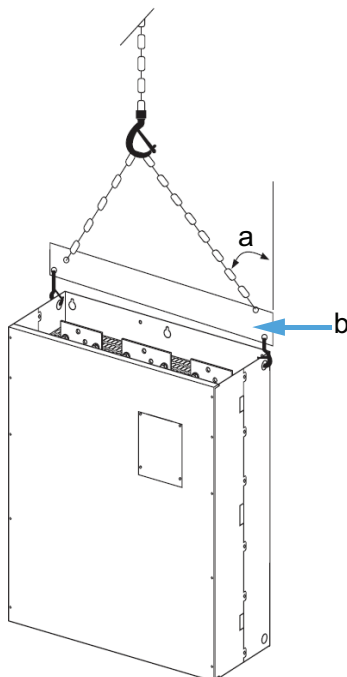
TOPPLING, SWINGING, OR FALLING EQUIPMENT

- Take all measures necessary to keep the equipment from swinging, toppling and falling.
- Follow the instructions provided to remove the equipment from the packaging and to mount it at its final position.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

See the procedure for hoisting the references from ATS480C41Y to ATS480M12Y:

Step	Action
1	Lift the soft starter by means of a hoist by using the handling lugs of the soft starter to fasten the lifting equipment. The lifting bar is not supplied.
2	Keep the soft starter suspended by means of appropriate equipment until it is securely fastened in the final installation position.
3	Move the soft starter to the final installation or on the back of the enclosure in accordance with the instructions given in this document.



- a: 45° maximum
- b: Lifting bar

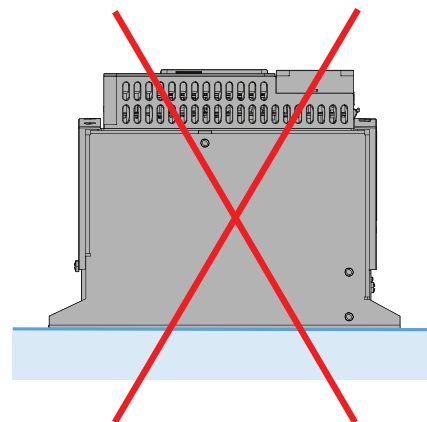
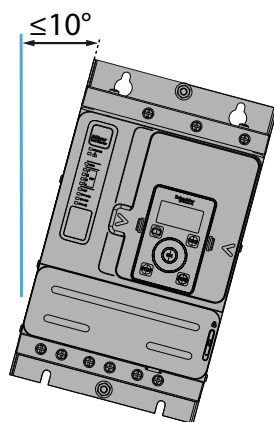
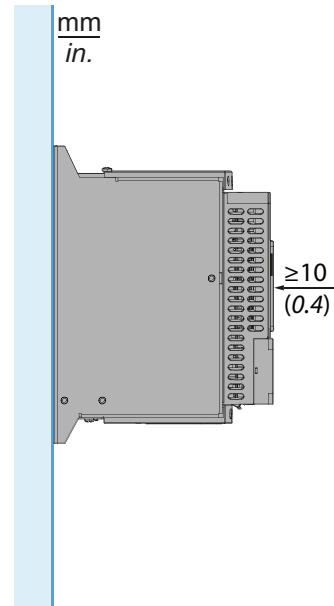
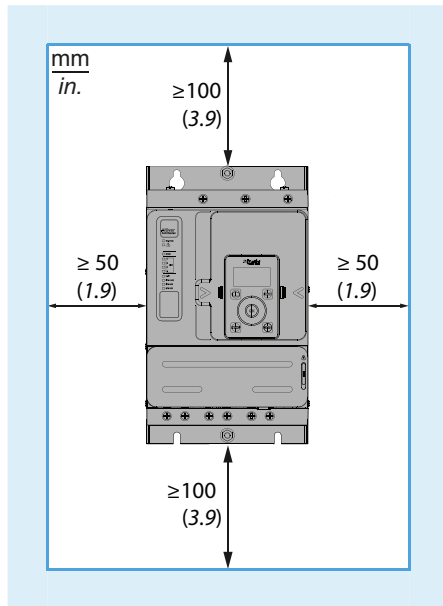
Clearances



The ATS48 and ATS480 have the same:

- Minimum clearances
- Maximum angle

No change is required.



Mounting

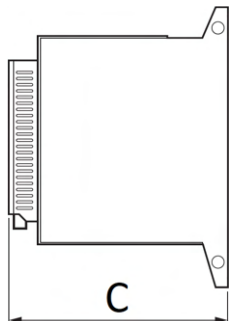


- The ATS480 use the same mounting plan as the ATS48.
- Use the same mounting holes from the existing ATS48 installation.
- Hole diameters, positions and mounting screws are identical.

Dimension Differences



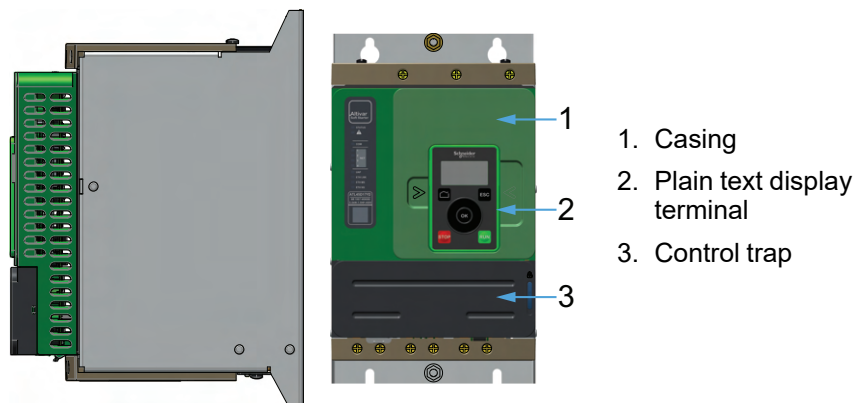
- The ATS480 and ATS48 have the same width and height.
- The ATS480 is deeper than the ATS48 (dimension “c” on the figure).
- Refer to the following table for the differences in depth.



References	ATS480 depth (dimension “c”) mm (inch)	Depth difference with ATS48 mm (inch)
ATS480D17Y...D47Y	203 (8)	+13 (0.51)
ATS480D62Y...C11Y	247 (9.72)	+12 (0.47)
ATS480C14Y...C17Y	272 (10.7)	+7 (0.27)
ATS480C21Y...C32Y	277 (10.9)	+12 (0.47)
ATS480C41Y...C66Y	314 (12.3)	+14 (0.55)
ATS480C79Y...M12Y	329 (12.95)	+14 (0.55)

Refer to the following table for possibilities to reduce the depth difference between the ATS48 and the ATS480.

Removable parts for depth reduction	Depth reduction if removed mm (inch)
Plain text display terminal	0.5 (0.019)
Plain text display terminal + control trap	3 (0.11)
Plain text display terminal + control trap + casing	5 (0.19)



NOTE: To assure an IP20 protection degree on the front with the ATS480D17Y...C11Y it is necessary to keep the casing.

ATS480 Remote Mounting Kits

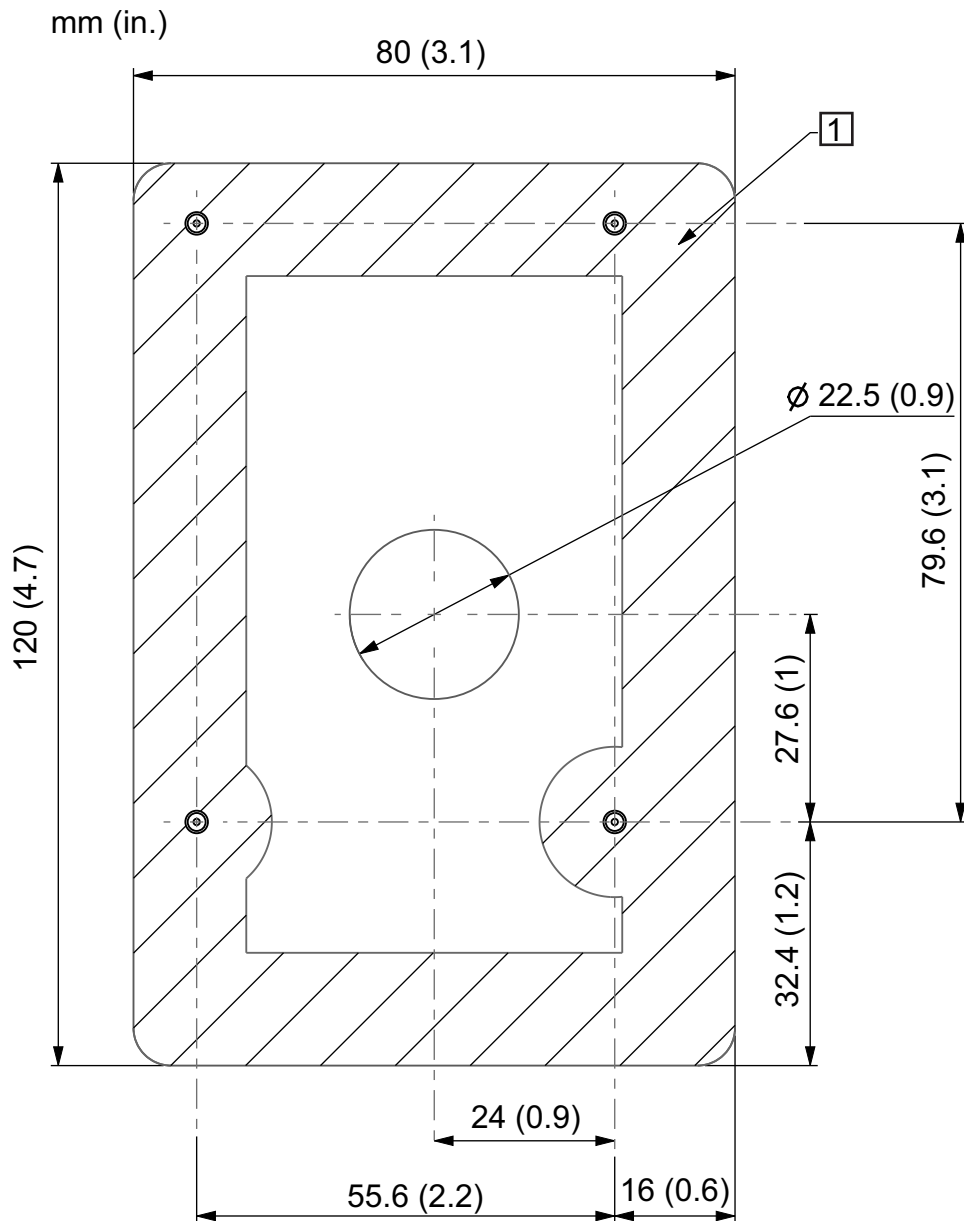


The drilling plans of the remote mounting kits for the Plain Text and Graphic Display Terminals are not compatible with the ATS48 door mounting kit drilling plan:

- The number and diameter of mounting holes is different
- The remote mounting kits for ATS480 are 1.5mm deeper than the ATS48 mounting kits

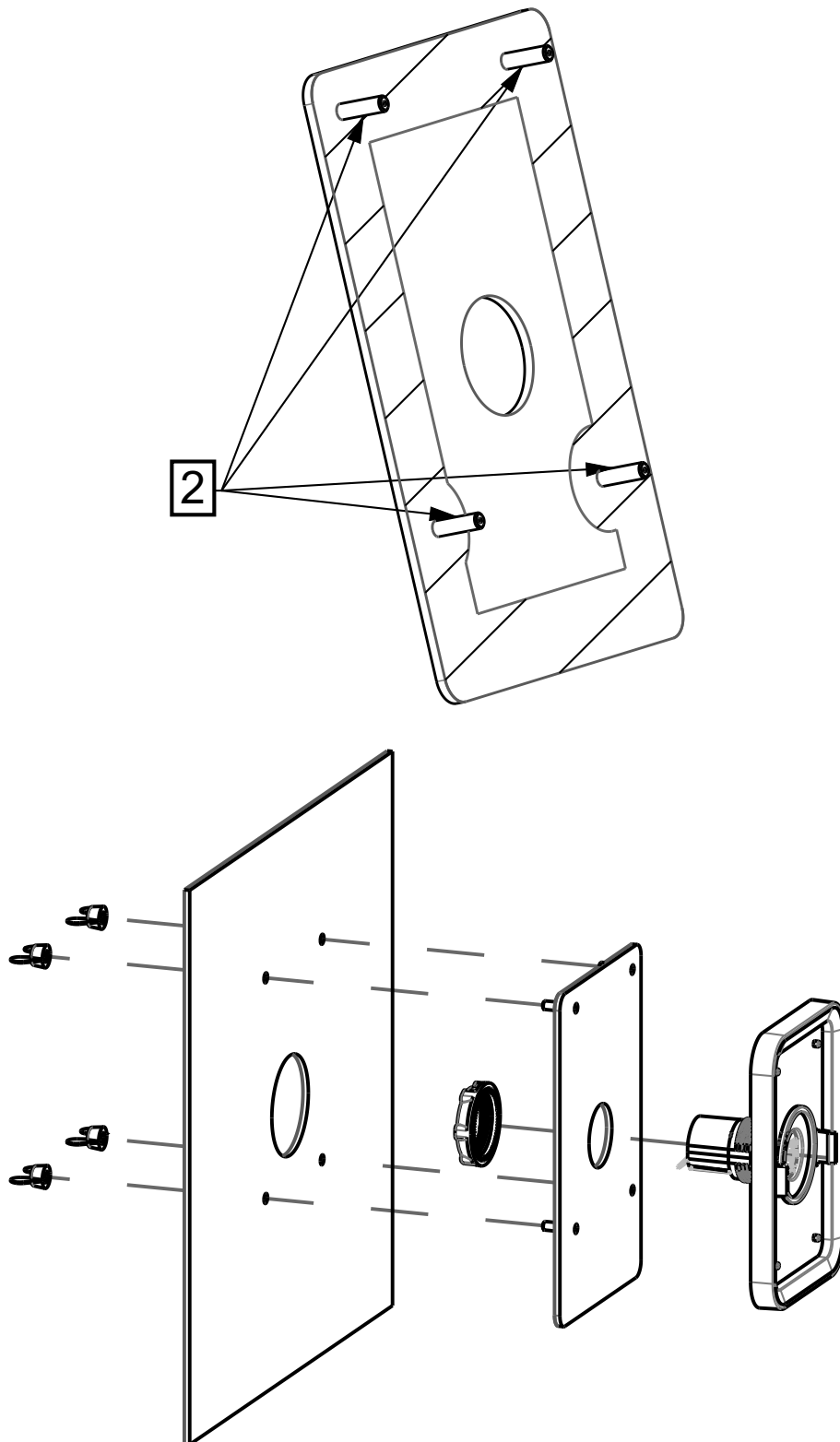
Refer to the following plan to create a support for the Plain Text Terminal and Graphic Display Terminal remote mounting kits.

1 Apply gasket for sealing



Scale 1:1

- 2 4 crimped stud FH M3 length 12mm (0.47 in.)



Wiring

What's in This Part

Power and Ground Wiring 37

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Power and Ground Wiring



- The ATS480 supply mains and ground wiring is identical to ATS48. The supply mains terminals denomination is identical between the two offers.
- The ATS48 electrical coordination elements, protections and contactors can be reused with the ATS480.

Identify the position of the mains supply cables on the ATS48 and identically connect the mains supply of the ATS480 with the same cables.

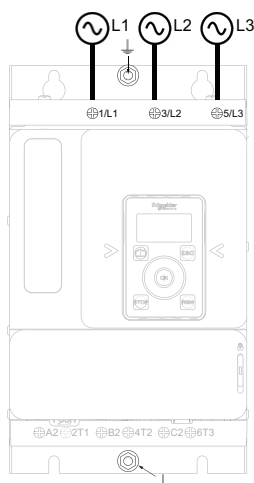
⚠ WARNING

UNANTICIPATED EQUIPMENT OPERATION

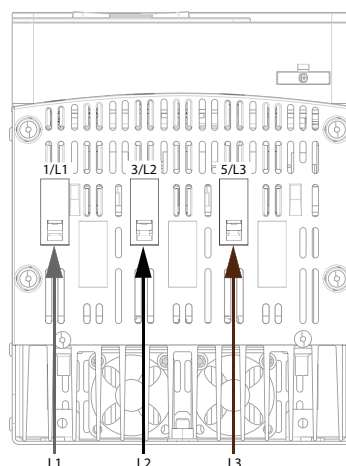
The ATS480 wiring must be identical to the ATS48 wiring in order to avoid wrong direction of the motor rotation.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

ATS480 front view



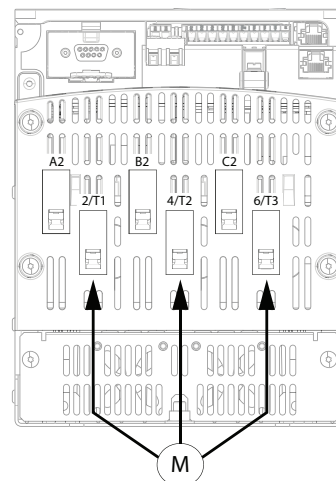
ATS480 top view



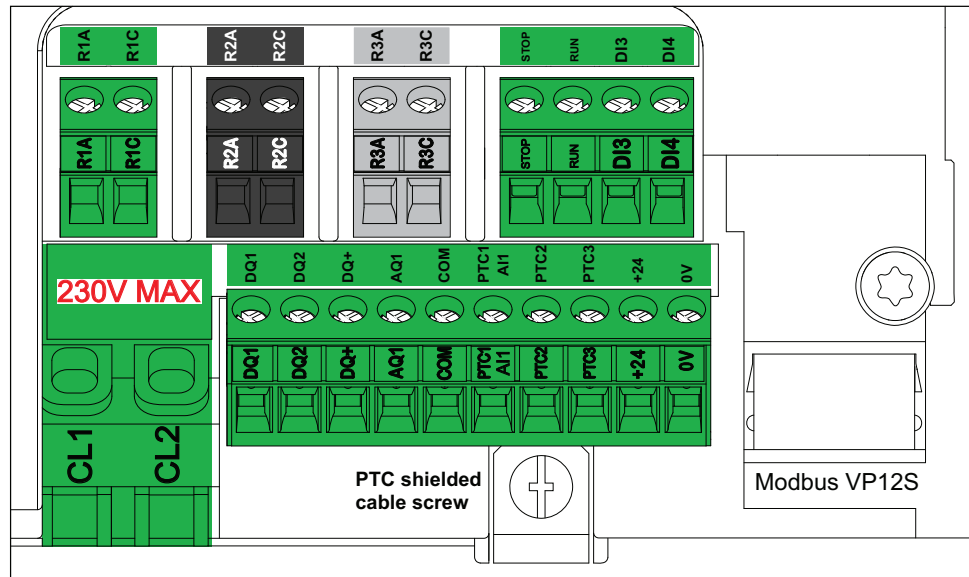
Connection characteristics

References	Tightening torque			
	Power connections		Ground	
	N·m	lbf.in	N·m	lbf.in
ATS480D17Y...D47Y	3	26	1.7	15
ATS480D62Y...C11Y	10	89	3	26
ATS480C14Y...C17Y	34	300	4.5	40
ATS480C21Y...C32Y	34	300	24	212
ATS480C41Y...C66Y	57	500		
ATS480C79Y...M12Y	57	500		

ATS480 bottom view



Layout And Characteristics Of Control Terminals



The control terminals are installed with one-way plug-in connectors and can be unplugged during the wiring.



The ATS480 control is supplied in 110...230 Vac +10% – 15%, 50/60 Hz ONLY.

- The ATS48●●●Y control block is supplied with 110...230 Vac. If the previous product is an ATS48●●●Y no change is required for the ATS480●●●Y.
- The ATS48●●●Q control block is supplied with 220...415 Vac. If the previous product is an ATS48●●●Q **you must adapt the supply voltage to 110...230 Vac** for the ATS480●●●Y.

You can use an existing 230 Vac supply source or a transformer to adapt the tension to 110 – 230 Vac.

NOTICE

INCORRECT VOLTAGE

- Supply the control supply terminals CL1 / CL2 within a range of 110...230 Vac only
- In case of migration from ATS48●●●Q to ATS480●●●Y, adapt the control supply transformer

Failure to follow these instructions can result in equipment damage.

Refer to this table to select the upstream electrical protection of CL1 / CL2:

References	Control apparent power (VA)
ATS480D17Y...D22Y	60
ATS480D32Y...C17Y	90
ATS480C21Y...C41Y	106
ATS480C48Y...C66Y	125
ATS480C79Y...M12Y	200



The ATS480 output relays allow for a maximum voltage of 230 Vac instead of 400 Vac on ATS48.

ATS48 terminals name	ATS480 terminals name	Description	Differences between ATS48 and ATS480
CL1	Same as ATS48	Control power supply	110...250 Vac, -15%...+10% Remove the sticker
CL2			
R1A	Same as ATS48	Programmable NO relay R1 – Assigned to Operating state Fault by default	Output relays allow for 230Vac instead of 400Vac on ATS48.
R1C			
R2A	Same as ATS48	NO relay R2 – Assigned to End of starting	
R2C			
R3A	Same as ATS48	Programmable NO relay R3	
R3C			
STOP	Same as ATS48	Digital Input 1 – Assigned to STOP	–
RUN		Digital Input 2 – Assigned to RUN	
LI3	DI3	Digital Input 3	
LI4	DI4	Digital Input 4	
COM	Same as ATS48	I/O common	–
LO+	DQ+	Digital output supply	–
LO1	DQ1	Programmable digital output 1	
LO2	DQ2	Programmable digital output 2	
AO1	AQ1	Programmable analog output 1	–
PTC1	PTC1/AI1	Motor Thermal sensor connection	+1 PTC terminal
PTC2	PTC2		
Not present	PTC3		
+24	Same as ATS48	Output: logic power supply / Input: device bloc control supply	–
Not present	0V	0V control	New terminal
RJ45 Modbus	Modbus VP12S	RS 485 Modbus	–

Control Terminals Wiring



- Reuse the wires of the ATS48 control terminals for the ATS480 control terminals.
- Wire the ATS480 control terminals identically to the ATS48 control terminals.
- The maximum connecting capacity and tightening torque are the same between ATS48 and ATS480 control terminals.

NOTE: The wires of the control part are 4.5 cm longer for the ATS480 references from ATS480D17Y to ATS480C17Y.

Maximum tightening torque	Minimum wires section		Maximum connection capacity
	Except relays	Relays	
N.m (lbf.in)	mm ² (AWG)		mm ² (AWG)
0.5 (4.4)	0.5 (20)	0.75 (18)	1.5 (15)

Refer to the following table of correspondence between the ATS48 to the ATS480 control terminals:

ATS48 terminals to unwire	Equivalent ATS480 terminals to wire
CL1	CL1
CL2	CL2
R1A	R1A
R1C	R1C
R2A	R2A
R2C	R2C
R3A	R3A
R3C	R3C
STOP	STOP
RUN	RUN
LI3	DI3
LI4	DI4
COM	COM
+24	+24
LO+	DQ+
LO1	DQ1
LO2	DQ2
AO1	AQ1
PTC1	PTC1 AI1
PTC2	PTC2
RJ45 Modbus	Modbus VP12S